

F24-101-D-Fabricae

Project Team

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Chapter 1

Introduction

In today's technology-centric world, emerging fashion designers face a growing need to maximize their resources, achieve valuable outcomes and market their skills. Recognizing these issues, our web application, Fabricae, will provide a complete platform to help independent and small-scale fashion designers overcome these resource constraints and simplify their design process. Our project helps designers create and showcase their designs in easy-to-follow steps. Firstly, designers can either generate a textile motif from scratch by using AI prompts or they can upload their own sketches which would be converted to 2D motifs. Using these motifs, they would create and customize their design pattern. To view this design pattern on actual garments, the designers can translate them onto custom 3D mockups where further 3D embellishments can also be added. The professional portfolio creation feature, which offers customizable templates, allows designers to showcase their creativity effectively. By displaying their portfolios on the platform, designers can connect with larger companies, opening doors to new opportunities and help in the advancement of their careers.

1.1 Problem Statement

Our platform enables designers to create unique and high-quality design patterns. Designers can create these patterns either by using descriptive prompts or by digitalizing their own pencil sketches without needing any expensive tools or a lot of time. Once the user will have the patterns, he can customize a pattern of his choice using our customization tool. Additionally, we simplify the design process by offering 3D mockups where designers can visualize their final product and would be able to add 3D embellishments on top (such as buttons and laces). This allows them to see the potential in their final designs without having to commit to any more resources. Beyond design, our platform also acts as a networking place where users can build and showcase their portfolios using

our customizable templates. This feature will help them connect with the industry professionals and bridge the gap between the big fashion companies and independent designers, opening doors for new opportunities and collaborations.

1.2 Scope

Fabricae is a web application that will enable designers to create unique textile patterns. The user will begin by signing into their account. They will then have two options: create patterns through descriptive prompts or create patterns by changing their uploaded pencil sketches into digital images. Next, the user can use these patterns to customize it from scratch using our editing tool. This tool will enable the user to customize their pattern using the options to add filters, apply color changes, adjust alignment and rotate the motifs to create a pleasing pattern. When the pattern has been created, the user can then check their final output by displaying it on a few custom 3D clothing mock-ups (both eastern and western i.e. shirt, pant, dress, shalwar kameez, long shirt). Furthermore, the user can add 3D embellishments (buttons and laces) to the model. The user may have the option to save their created patterns in our database and then in future use these patterns while creating their portfolios. Moreover, the user can create their own portfolio using our 2-3 portfolio templates and then publish them on our platform. If a designer brand wants to approach the user because of their portfolio, they can do so using our chat feature.

1.2.1 Prompt-based Textile Pattern Generation

This module allows users to type in descriptive text prompts to generate design patterns. Users can then select more than one pattern for further editing. The combination of the selected patterns will then be used in our editing tool where he can create a final pattern of his choice.

Features:

1. Dataset creation (collecting and captioning the images)
2. Pattern generation using textual descriptive prompts.
3. History feature to view the previous prompt-based motif conversions

1.2.2 Sketch-to-2D Motif Conversion

In this module, users can upload sketches of their textile patterns which will be converted to 2D digital design. After this, the user can select it for further customization and editing.

Features:

1. Dataset creation
2. Image upload functionality for sketches
3. Pencil sketch to digital image conversion
4. History feature to view the previous sketch-to-2D motif conversions

1.2.3 Pattern Creation and Editing

This module provides tools for designers to customize their textile patterns. The user will be given the option to drag and drop the different patterns he selected from the pattern generation module onto a blank white canvas. He will then create a final pattern using the motifs and the customization tool which includes options to apply filters, color changes and alignment adjustment.

Features:

1. Converting generated motifs to editable objects
2. Drag, drop, rotate and scale your selected motifs
3. Change pattern colors
4. Apply filters on patterns

1.2.4 3D Visualization and Embellishment

Users can wrap their created patterns onto 3D models of clothing items and apply embellishments like buttons and laces on top of it. In this module, the user would have the option to view his design patterns on a 3D model and view it from all sides.

Features:

1. Create 3D clothing mock-ups on blender
2. Drag and position the 3D embellishments on the 3D mock-ups
3. 360-degree view of 3D mock-ups

4. Applying created patterns to 3D clothing models
5. Export designs in format such as PNG, glTF and OBJ
6. Add watermark to protect ownership of designs

1.2.5 Portfolio Creation

This module will allow users to create fashion portfolios using our pre-defined templates. The users can upload their created designs, add descriptions, images and videos so that they can have a personalized portfolio creation experience.

Features:

1. Customizing the pre-designed Portfolio templates
2. Upload functionality for designs on the templates

1.2.6 Portfolio Publishing and Networking

Users can publish their portfolios on our platform. This way the big fashion companies or the industry professionals can view these portfolios and contact them through a chat box. Over here the users can also share their portfolios with social media options for further networking and recognition.

Features:

1. Portfolio Publishing
2. Messaging through chat box
3. Portfolio sharing on media platforms
4. Message alerts on portfolio engagement

1.3 User Classes and Characteristics

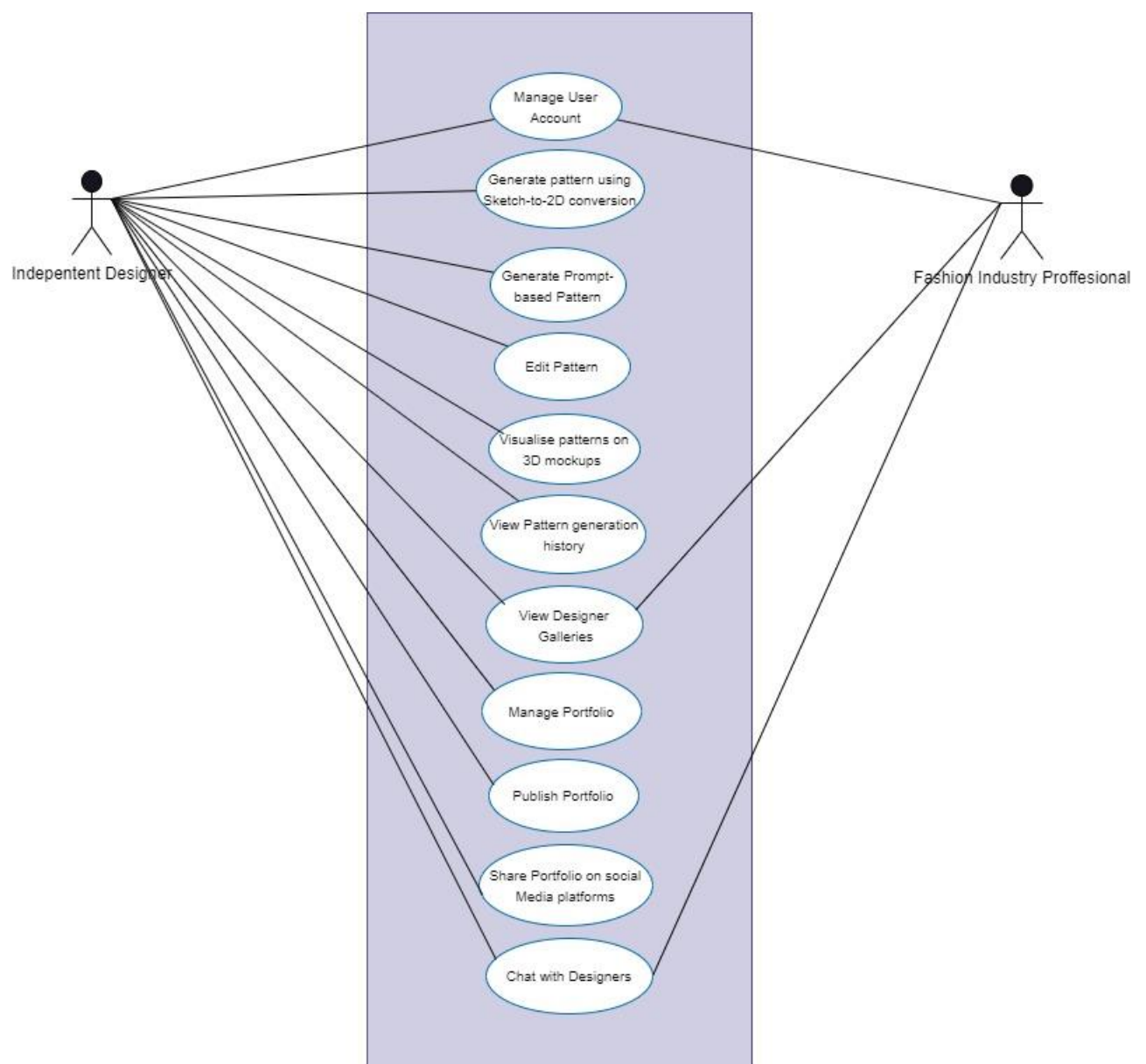
User class	Description
Fashion Designer	A fashion designer will use this app to generate different textile patterns either by using descriptive prompts or by digitalizing their pencil sketches. He may have limited technical skills but he has great ideas to implement. The designer has the ability to make clear and pleasing pencil sketches. The designer likes to work independently or in a small team. He can generate as many patterns as he wants and then save them as a history for future use. He is also looking forward to creating a strong portfolio that could gain the attention of industry
Industry Professional	The industry professional may not be interested in the entire design process but he may want to look at the final design outputs of different designers. He wants to easily navigate through the design feed and find the best portfolios and designs he is looking for. He may also want to chat with the designer in order to further collaborate on a future project or hire him.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use Cases



2.1.1 High Level Use Cases

Use case ID:	UC-01
Use Case:	Manage User Account
Actors:	Designer, Industry Professional
Type:	Primary
Description	Users can sign up using personal details or third-party platforms, log in, edit their account information, and delete their account if needed.

Use case ID:	UC-02
Use Case:	Generate Prompt-based pattern
Actors:	Designer
Type:	Primary
Description	Designers can input descriptive prompts to generate multiple pattern options.

Use case ID:	UC-03
Use Case:	Generate Sketch-to-2D pattern
Actors:	Designer
Type:	Primary
Description	Designer can upload his own sketch. The system will convert and generate digital patterns of the uploaded sketch.

Use case ID:	UC-04
Use Case:	Edit Pattern
Actors:	Designer
Type:	Primary
Description	Designers can use our tool to drag and drop patterns onto a blank canvas and apply different filters, adjust alignments, change colors, and crop pattern to create his own design.

Use case ID:	UC-05
Use Case:	Visualize patterns on 3D-mockups
Actors:	Designer
Type:	Primary
Description	Designers can showcase their patterns on 3D mockups and position 3D embellishments on it. They can also export their design in various formats.

Use case ID:	UC-06
Use Case:	Manage Portfolio
Actors:	Designer
Type:	Primary
Description	Designers can create portfolios using our pre-defined templates. They can also view, delete and edit them if needed.

Use case ID:	UC-07
Use Case:	View Design Galleries
Actors:	Designer, Industry Professional
Type:	Secondary
Description	Users can explore design galleries which includes portfolios and pattern designs created by independent designers

Use case ID:	UC-08
Use Case:	Chat with Designers
Actors:	Designer, Industry Professional
Type:	Secondary
Description	Fashion industry professionals can engage with the designers through an integrated chat feature. Designers can also share their portfolios on social mediaplatforms.

Use case ID:	UC-09
Use Case:	View Pattern Generation History
Actors:	Designer
Type:	Secondary
Description	Designers can view and reuse old patterns from both prompt-based, 2D sketches conversions and edited patterns.

Use case ID	UC-11
Use Case	Share Portfolio on Social Media
Actors	Designer
Type	Secondary
Description	Designers can use our platform and link their social media accounts (e.g. Facebook, Instagram) for sharing portfolios

Use case ID	UC-10
Use Case	Publish Portfolio
Actors	Designer
Type	Primary
Description	Designers can publish their created portfolios on Fabricae platform in order to network with the big fashion brands

2.1.2 Extended Use Cases

Name	Manage User Account	
Scope	Account Management	
Level	User goal	
Actor	Primary Actor: Independent Designer Secondary Actor: Industry Professionals	
Stakeholders and interest	Designers: They want to create their accounts so they can create, customize their designs and showcase them in their portfolios to fashion brands Industry Professionals: They want to discover new talent by viewing independent fashion designer's work and be able to easily converse with them Fabricae Platform: They want to ensure a secure and smooth user experience to keep users.	
precondition	• Users must have internet access to access the platform • Users must have valid email addresses for account creation	
Success guarantee	The account management operations (create, log in, edit, delete) are performed successfully without errors.	
Main success scenario	User registration:	
	User Action	System Response
	The user clicks on the 'Sign Up' button and navigates to sign up page	
		The system displays all the required input fields on the sign-up page to the user.
	The user fills out the form with their email, first name, last name, and password.	
		The system validates the user's required information according to already set name and password requirements
	The user submits the form by clicking on the sign-up button	

	The system validates the input and creates a new account.
	The system sends an email for confirmation to the user.

User Login:

The user navigates to the login page	
	The system displays the login screen to him
The user inputs their email and password	
	The system validates their inputs
The user clicks on 'login' button	
	The system validates the user's credentials and allows access to them
The user is navigated to the homepage	

Edit account details:

The user navigates to the profile settings	
	The system displays the profile page with all their current personal data
The user updates their name,email or password.	
	The system provides real time user validation as they type.
The user clicks on the save changes button.	
	The system validates the user data and updates the account information.

	Delete Account:	
	The user navigates to profile settings page	
		The system displays the profile settings page to them
	The user clicks on the ‘delete account’ option	
		The system prompts the user to confirm the deletion
	The user clicks on delete	
		The system deletes the user account and its data
	The user gets a message ‘Account successfully created’ displayed to them	
	Logout:	
	The user clicks on the logout option on the homepage	
		The system logs out the user
	The user is redirected to homepage and they view that they no longer have access to their account related information	
Extensions	Success: For signup, user can use third-party platforms • The user clicks the chosen platform (e.g., Google). • The user is redirected to the third-party authentication page. • The user enters their credentials and authorizes the platform to share information with the service. Failures: Error Handling:	

	1. Registration Errors: ○ In the case of email being already in use, the system displays the message to the user to log in instead ○ If the password does not meet the set input validations, the system displays an error message. 2. Login Errors: ○ If the user enters an incorrect email or password, the system informs the user and provides an option for password resetting, if needed. 3. Editing Errors: ○ If the user tries to submit an empty name or invalid email format, the system displays an error message. 4. Account Deletion Confirmation: ○ If the system asks the user for confirmation of account deletion and the user does not choose to confirm that, the system returns to the account settings page without the removal of account 5. Session Expiration: ○ If the user's session times out, the system prompts them to log in again to continue.

Name	Generate Prompt-Based Pattern
Scope	Pattern Generation
Level	User goal
Actor	Primary Actor: Independent Designer Secondary Actor: None
Stakeholders and interest	Designers: • Interest: To be able to create unique patterns using descriptive prompts for their portfolios or projects Fabricae Platform: • Interest: To provide designers with tools that enhance creativity and productivity

Pre-Condition	- The designer must have stable internet connection - The designer must have a valid user account and be logged into the app - The designer must have access to prompt-based generation feature												
Success guarantee	- The system generated multiple patterns based on the user’s prompt - The user can select one or more pattern - The system maintains pattern history												
Main success scenario	<table><tr><th>Actor Action</th><th>System Response</th></tr><tr><td>1. Designer enters a descriptive prompt</td><td></td></tr><tr><td></td><td>2.System processes the prompt and generates multiple patterns</td></tr><tr><td>3.Designer selects one or more patterns</td><td></td></tr><tr><td></td><td>4. The system highlights the selected pattern and ‘lets customize’ button.</td></tr><tr><td>5.The user clicks on ‘lets customize’ button</td><td></td></tr></table>	Actor Action	System Response	1. Designer enters a descriptive prompt			2.System processes the prompt and generates multiple patterns	3.Designer selects one or more patterns			4. The system highlights the selected pattern and ‘lets customize’ button.	5.The user clicks on ‘lets customize’ button	
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	2.System processes the prompt and generates multiple patterns												
3.Designer selects one or more patterns													
	4. The system highlights the selected pattern and ‘lets customize’ button.												
5.The user clicks on ‘lets customize’ button													
extensions	Prompt Validation Failure: Step: If the designer enters an invalid prompt (e.g., too short, less descriptive). System Response: The system displays an error message (e.g., "Please enter a more detailed prompt.") and asks the designer to enter the prompt again. No patterns Generated: Step: The system fails to generate patterns because of technical issues. System Response: The system displays an error message (e.g., "Unable to generate patterns right now. Please try later."). Pattern Saving Failures: Step: The system attempts to save customized patterns, but the saving process fails maybe due to internet issues.												

	• System Response: The system displays the message to the designer (e.g., "Failed to save patterns. Please check your connection and try again.").
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Name	Generate Sketch-to-2D Pattern	
Scope	Pattern Generation	
Level	User goal	
Actor	Primary Actor: Independent Designer Secondary Actor: None	
Stakeholders and interest	Designer: To be able to create digital patterns by using their own pencil sketches for their portfolios or projects so that they can translate their creativity Fabricae Platform: Interest: To provide designers with tools that enhance creativity and productivity	
precondition	- The designer must have stable internet connection - The designer must have a valid user account and be logged into the app - The designer must have access to sketch to 2D pattern feature. - The sketch is in an acceptable file format (e.g., PNG, JPEG).	
Success guarantee	The designer can select the generated patterns.	
Main success scenario		
	Actor Action	System Response
	1.The designer uploads a sketch	
		2.The system allows the image upload and displays it to the user
	3.The designer checks if the uploaded sketch is correctly uploaded	

		4.The system processes the uploaded sketch and generates pattern.
		5. The system highlights the ‘lets customize’ button.
	6.The designer clicks on ‘lets customize’ button	
extensions	Upload Validation Failure: • Step: If the designer uploads a file in an unsupported format. • System Response: The system displays an error message (e.g., "Unsupported file format. Please upload a PNG or JPEG file."). Processing Failure: • Step: The system fails to process the uploaded sketch due to technical issues. • System Response: The system displays an error message (e.g., "Unable to process the sketch at this time. Please try again later."). Pattern Saving Failure: • Step: The system attempts to save customized patterns, but the save operation fails (e.g., due to internet connectivity issues). • System Response: The system notifies the designer (e.g., "Failed to save patterns. Please check your connection and try again.").	

Name	Edit Pattern
Scope	Pattern Creation and Editing
Level	User goal
Actor	Primary Actor: Independent Designer Secondary Actor: None
Stakeholders and interest	Designers:

	• Interest: To bring his authentic designs to life by creating and customizing his pre-generated patterns using the available editing tool options																						
precondition	• The designer must have stable internet connection • The designer must have a valid user account and be logged into the app • The designer must have access to the editing tool • The designer must have access to his selected generated patterns from the earlier pattern generation step																						
Success guarantee	• The designer can create and customize patterns by using available tool options • The system saves the final design created by the designer																						
Main success scenario	<table> <tr> <th>Actor Action</th><th>System Response</th></tr> <tr> <td>1.The user navigates to the editing tool and selects a pattern from their selected patterns</td><td></td></tr> <tr> <td></td><td>2.The system highlights the selected pattern</td></tr> <tr> <td>3.The user drags and drops the pattern onto the canvas</td><td></td></tr> <tr> <td></td><td>4.The system places the pattern on the canvas</td></tr> <tr> <td>5.The designer choose a filter to apply to the pattern.</td><td></td></tr> <tr> <td></td><td>6.The system applies the selected filter, displaying the modified pattern on canvas</td></tr> <tr> <td>7.The user adjusts alignment of the pattern (e.g flip, rotate)</td><td></td></tr> <tr> <td></td><td>8.The system updates the pattern's position and orientation</td></tr> <tr> <td>9.The user selects a color palette and changes the color of the selected parts of the pattern</td><td></td></tr> <tr> <td></td><td>10.The system updates the pattern's colors</td></tr> </table>	Actor Action	System Response	1.The user navigates to the editing tool and selects a pattern from their selected patterns			2.The system highlights the selected pattern	3.The user drags and drops the pattern onto the canvas			4.The system places the pattern on the canvas	5.The designer choose a filter to apply to the pattern.			6.The system applies the selected filter, displaying the modified pattern on canvas	7.The user adjusts alignment of the pattern (e.g flip, rotate)			8.The system updates the pattern's position and orientation	9.The user selects a color palette and changes the color of the selected parts of the pattern			10.The system updates the pattern's colors
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	10.The system updates the pattern's colors																						

	11.The user selects the cropping tool and crops the pattern to fit a specific area	
		12.The system shows the cropped version of the pattern on canvas
	13.The designer clicks on 'visualize' button	
		14.The system takes the user to the 3D mock-up visualization page
Extensions	Alternative success scenario: 13. Designer selects the format he wants to export the design in (e.g png, obj) and exports the design 14.The system generates the exported file and displays a confirmation message "Your design has been exported" Pattern Selection Failure: • Step: If the designer tries to select a pattern that is unavailable or corrupted. • System Response: The system displays an error message (e.g., "Unable to load the selected pattern at the moment. Please choose another one."). Invalid Drag-and-Drop: • Step: If the designer tries to drag the pattern outside the canvas area. • System Response: The system restricts the movement and displays message that the pattern cannot be placed outside the canvas. Filter Application Error: • Step: The designer applies a filter, but the system encounters a processing issue. • System Response: The system displays an error message (e.g., "Unable to apply the selected filter. Please try again."). Alignment Tool Failure: • Step: If the alignment tool fails to adjust the pattern properly.	

	• System Response: The system notifies the designer (e.g., "Alignment adjustment failed. Please reload the pattern and try again."). Colour Change Error: • Step: The designer tries to change the color, but the color picker does not work. • System Response: The system displays an error (e.g., "Unable to change color at this time. Please try again.") and enables the designer to retry. Cropping Error: • Step: The designer attempts to crop the pattern but encounters an issue (e.g., the crop area is too small). • System Response: The system provides a warning (e.g., "Crop area is too small. Please select a larger area.") and prevents the action. Saving Failure: • Step: The system tries to save the edited pattern, but a connection issue occurs. • System Response: The system notifies the designer (e.g., "Failed to save the design. Please check your connection and try again.").
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Name	Publish Patterns
Scope	Pattern Creation
Level	User goal
Actor	Primary Actor: Independent Designer, Industry professional Secondary Actor: None
Stakeholders and interest	Designers: • Interest: To explore the gallery and get inspired by other designer's work Industry Professionals: • Interest: To discover new talent by exploring unique designs and connect with the designers who made them

	Fabricae Platform: Interest: To gain platform exposure and attract designers by showing the type of designs that can be created by using our platform	
Precondition	- The designer must be logged into the platform with a stable internet connection - Independent designers must have created designs.	
Success guarantee	The user will be able to successfully publish his/her patterns on the platform.	
Main success scenario	Actor Action	System Response
	1.The user selects the publish button on the Pattern Editing screen.	
		2.The system prompts a publish form to the user with fields like Design title and description.
	3.The user inputs the relevant information about the design post	
		3.The system Publishes the pattern to the designer’s design gallery.
	4. The user can now view the published design in the design gallery.	
Extensions	Failed to Publish: Step: If the system fails to publish to the design gallery due to a technical issue. System Response: The system displays an error message “Cannot Publish your Design, Try again Later. Thank you”	

Name	Visualize patterns on 3D-mockups
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Scope	3D Pattern Visualization	
Level	User goal	
Actor	Primary Actor: Independent Designer Secondary Actor: None	
Stakeholders and interest	Designers: Interest: To visualize their created patterns on 3D mockups and adding embellishments to it so that they can see the final outcome is similar to their creative vision Fabricae Platform: Interest: To provide a platform offering visualization tool for the designers which enhances user engagement and product offering.	
precondition	- The designer must have a stable internet connection and must be logged into the app - The designer must have a created pattern from editing pattern step - The designer must have access to the visualization tool	
Success guarantee	- The designer can apply patterns to 3D mock-ups and apply the 3D embellishments on top - The system saves the final mock-up and allows the designer to export it in the desired format	
Main success scenario	Actor Action	System Response
	1.The user navigates to the 3D visualization tool and selects a mockup style (e.g dress, shirt)	
		2.The system loads the selected 3D mockup style and displays it on the screen
	3.The user selects pattern and drops it on the 3D mockup	
		4.The system applies the selected pattern on the mockup
	5.The user selects embellishments (e.g buttons,laces) and positions them on the mockup	

		6.The system displays the embellishment on the selected part of the mockup
	7.The designer rotates ad zooms the mockup to view it from all the different angles	
		8.The system renders the model and shows it from different angles
	9.The designer selects an export format (e.g png,obj) and exports the design	
		10.The system generates the exported file and displays a confirmation message “ your design has been exported”
Extensions	Mockup Selection Failure: • Step: The designer selects a mockup that is not rendering • System Response: An error message “Unable to load the selected mockup, please select another option” is shown to the user Pattern Application Error: • Step: The user tries placing the pattern on the model but there is a problem with system’s processing • System Response: An error is displayed by the system “Unable to apply the selected pattern, Kindly Try again Later!” to the user Invalid Embellishment Positioning: • Step: If the designer tries to place the embellishment in an unsupported area • System Response: The system responds by placing restrictions and displaying an error message “ Embellishment cannot be placed in the selected area” Export Failure: • Step: The designer tries to export the design, but a connection issue or system error occurs. • System Response: The system notifies the designer (e.g., "Failed to export the design. Please check your connection and try again.").	

Name	Manage Portfolio	
Scope	Portfolio Publishing and Networking	
Level	User goal	
Actor	Primary Actor: Independent Designer Secondary Actor: None	
Stakeholders and interest	Designers: Interest: To create, modify, and manage portfolios adequately, allowing designers to present their work professionally Business Owners: Interest: To enable more designers to use the tool and platform for their work professionally and make the user experience accessible and convenient	
precondition	- The designer must have stable internet connection and be logged into the app - The designer must have access to the pre-defined portfolio templates - The designer must have access to the content that he wants to add to his portfolio (pictures, designs, videos etc)	
Success guarantee	The designer can create, update, view, or delete their portfolios according to their needs, and in case of any changes, the updates be saved and reflected as is in their profile.	
Main success scenario	Create Portfolio:	
	Actor Action	System Response
	1.The user navigates to manage portfolio page	
		2.The system displays a list of existing portfolios and displays ‘create portfolio’ button
	3.The user clicks on the button	
		4.The system displays multiple pre-defined templates

	5.The user chooses a predefined template	
		6.The system displays the selected template with editable fields
	7.The user inputs his work details into the fields (e.g images, project details, videos)	
		8.The system saves the portfolio and shows a preview
	View Portfolio:	
	1.The user selects a portfolio of his that he wants to view.	
		2.The system displays the full content of the selected portfolio, including images and descriptions.
Edit Portfolio:		
	1.The designer selects a portfolio to edit and updates the content (e.g., changes images, text, or layout).	
		2.The system validates the edits made by the user, saves the updated data, and displays the success message to indicate that the changes are reflected properly
Delete Portfolio:		
	The designer selects a portfolio to delete.	
		The system asks for confirmation and, upon confirmation, deletes the portfolio and displays a success message.

Extensions	Incomplete Portfolio Information: • Step: If the designer attempts to create a portfolio without completely filling in all the required fields. • System Response: The system displays an error message (e.g., "Please complete all required fields before saving the portfolio."). Editing Failure: • Step: If the system encounters an error while saving edits made to a portfolio. • System Response: The system displays an error message (e.g., "Unable to save the changes made. Please try again."). Deletion Confirmation Denied: • Step: If upon the confirmation message by the system, the designer decides not to delete their portfolio • System Response: The system cancels the deletion and returns to the portfolio management page. Duplicate Portfolio Name: • Step: If the designer tries to create a new portfolio with the name as an existing one. • System Response: The system asks the designer to provide another name that is different (e.g., "A portfolio with this name already exists. Please choose a different name.").

Name	Publish Portfolio
Scope	Portfolio Creation and Publishing
Level	User goal
Actor	Primary Actor: Independent Designer Secondary Actor: None
Stakeholders and interest	Designers: • Interest: To publish their work in a professional manner and get noticed by big fashion brands. Fabricae Platform:

	• Interest: To help form connections between designers and fashion brands, helping in networking.										
precondition	• The designer must have stable internet connection • The designer must have a valid user account and be logged into the app • The designer must have minimum of one portfolio that is prepared to be published										
Success guarantee	The designer's portfolio is published on the Fabricae platform and it is available for viewing by other users or fashion brands										
Main success scenario	<table border="1"> <thead> <tr> <th>Actor Action</th><th>System Response</th></tr> </thead> <tbody> <tr> <td>1.The user navigates to the portfolio page</td><td></td></tr> <tr> <td></td><td>2.The system displays a list of portfolios he has created</td></tr> <tr> <td>3. The designer selects a portfolio and clicks the "Publish" button.</td><td></td></tr> <tr> <td></td><td>4.The system confirms that the portfolio is ready to be published and displays message “ Your portfolio has been published” to the user</td></tr> </tbody> </table>	Actor Action	System Response	1.The user navigates to the portfolio page			2.The system displays a list of portfolios he has created	3. The designer selects a portfolio and clicks the "Publish" button.			4.The system confirms that the portfolio is ready to be published and displays message “ Your portfolio has been published” to the user
Actor Action	System Response										
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3. The designer selects a portfolio and clicks the "Publish" button.											
	4.The system confirms that the portfolio is ready to be published and displays message “ Your portfolio has been published” to the user										
Extensions	Duplicate Portfolio Warning: • Step: If the designer attempts to republish the same portfolio • System Response: A message is displayed by the system to the user “This portfolio has already been published”										

Name	View Pattern Generation History
Scope	Pattern Creation
Level	User goal
Actor	Primary Actor: Independent Designer Secondary Actor: None

Stakeholders and interest	Designers: Interest: To be able to save time and effort when producing new designs by having access to previously developed patterns for reuse and or reference Fabricae Platform: Interest: To offer a practical and easily available functionality for pattern history that boosts user interaction and platform usage													
precondition	- The designer must have stable internet connection and be logged into the app. - The designer must have some generated or edited patterns made by the platform features.													
Success guarantee	The designer will be able to view all previously generated patterns and reuse them for future projects.													
Main success scenario	<table> <tr> <th>Actor Action</th> <th>System Response</th> </tr> <tr> <td>1.The user navigates to the pattern generation history page</td> <td></td> </tr> <tr> <td></td> <td>2.The system displays a list of previously generated patterns organized by date, type (prompt-based, 2D sketch conversion, edited).</td> </tr> <tr> <td>3. The designer selects a specific pattern from the history list to view more details.</td> <td></td> </tr> <tr> <td></td> <td>4. The system displays the pattern in detail, including generation type, modifications made, and associated metadata (e.g., date created).</td> </tr> <tr> <td>5.The designer chooses a pattern to reuse and selects the option to ‘Customize pattern’ for a new project</td> <td></td> </tr> </table>		Actor Action	System Response	1.The user navigates to the pattern generation history page			2.The system displays a list of previously generated patterns organized by date, type (prompt-based, 2D sketch conversion, edited).	3. The designer selects a specific pattern from the history list to view more details.			4. The system displays the pattern in detail, including generation type, modifications made, and associated metadata (e.g., date created).	5.The designer chooses a pattern to reuse and selects the option to ‘Customize pattern’ for a new project	
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		4. The system displays the pattern in detail, including generation type, modifications made, and associated metadata (e.g., date created).												
	5.The designer chooses a pattern to reuse and selects the option to ‘Customize pattern’ for a new project													

Extensions	No Patterns in History: • Step: If the designer has not yet created or modified any patterns • System Response: The system displays a message (e.g., "No pattern history available. Start creating patterns now.") with an option to navigate to the pattern generation tool. Search and Filter Patterns: • Step: The designer wants to search or filter patterns by type or date. • System Response: The system provides search and filtering options (e.g., by date, pattern type) to help the designer locate a specific pattern more easily. Pattern Reuse Failed: • Step: If the system fails to load the pattern for reuse due to a technical issue. • System Response: The system displays an error message (e.g., "Unable to load the selected pattern. Please try again later."). Pattern Deletion • Step: The designer chooses to delete a pattern from the history. • System Response: The system requests confirmation, and upon confirmation, deletes the pattern and displays a success message
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Name	View Design Galleries
Scope	Pattern Creation and Editing
Level	User goal
Actor	Primary Actor: Independent Designer, Industry professional Secondary Actor: None
Stakeholders and interest	Designers: • Interest: To explore the gallery and get inspired by other designer's work Industry Professionals: • Interest: To discover new talent by exploring unique designs and connect with the designers who made them Fabricae Platform:

	• Interest: To gain platform exposure and attract designers by showing the the type of designs that can be created by using our platform												
Precondition	• The designer must be logged into the platform with a stable internet connection • Independent designers must have designs that are published in order for them to be viewed by other designers and professionals												
Success guarantee	The user will be able to browse, filter, and interact with the design galleries effectively, discovering designs and portfolios of independent designers.												
Main success scenario	<table border="1"> <thead> <tr> <th>Actor Action</th><th>System Response</th></tr> </thead> <tbody> <tr> <td>1.The user goes to the explore page</td><td></td></tr> <tr> <td></td><td>2.The system displays a list of designs along with the designer and design information</td></tr> <tr> <td>3. The user applies filters (e.g., design type, creation date, designer name) or searches for a specific design.</td><td></td></tr> <tr> <td></td><td>4. The system updates the gallery according to the search/filter criteria.</td></tr> <tr> <td></td><td></td></tr> </tbody> </table>	Actor Action	System Response	1.The user goes to the explore page			2.The system displays a list of designs along with the designer and design information	3. The user applies filters (e.g., design type, creation date, designer name) or searches for a specific design.			4. The system updates the gallery according to the search/filter criteria.		
Actor Action	System Response												
1.The user goes to the explore page													
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3. The user applies filters (e.g., design type, creation date, designer name) or searches for a specific design.													
	4. The system updates the gallery according to the search/filter criteria.												
Extensions	No Designs Available in Gallery: • Step: If no designs are currently available in the searched/filter criteria. • System Response: The system displays a message (e.g., "No designs available in this category. Please try another category."). Failed to Load Gallery: • Step: If the system fails to load the gallery due to a technical issue. • System Response: An error message stating “ Cannot load Design Gallery. Try again Later” is displayed to the user												

Name	Chat with Designers
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Scope	Portfolio Publishing and Networking											
Level	User goal											
Actor	Primary Actor: Independent Designer, Industry professional Secondary Actor: None											
Stakeholders and interest	Designers: Interest: To network with the specialists in the industry, share portfolios and build connections for possible opportunities Industry Professionals: Interest: To easily connect with designers, discover new talent, discuss potential projects, and get more information about their work and portfolios. Fabricae Platform: Interest: To help bridge the gap between designers and industry professionals, increase platform engagement, and encourage potential collaborations.											
precondition	- Both the designer and the industry professional must be registered users on the Fabricae platform. - The chat feature must be enabled and accessible within the platform.											
Success guarantee	The Industry professionals and designers will be able to get in contact through chat, share details, and make meaningful connections.											
Main success scenario	<table><tr><th>Actor Action</th><th>System Response</th></tr><tr><td>1.The industry professional navigates to the Explore page and clicks on the ‘chat’ button under a specific design he likes</td><td></td></tr><tr><td></td><td>2.The system opens a chat window to the the professional, allowing them to start typing a message out and then hit send</td></tr><tr><td>3. The designer receives a chat notification and reads the message</td><td></td></tr><tr><td></td><td>4. The system notifies the industry professional of the new message</td></tr></table>		Actor Action	System Response	1.The industry professional navigates to the Explore page and clicks on the ‘chat’ button under a specific design he likes			2.The system opens a chat window to the the professional, allowing them to start typing a message out and then hit send	3. The designer receives a chat notification and reads the message			4. The system notifies the industry professional of the new message
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3. The designer receives a chat notification and reads the message												
	4. The system notifies the industry professional of the new message											

	<table border="1"> <tr> <td>5. The user reads the message and communication continues</td><td></td></tr> </table>	5. The user reads the message and communication continues	
5. The user reads the message and communication continues			
Extensions	Designer Not Available for Chat: • Step: If the designer is currently unavailable or offline. • System Response: The system displays an offline icon indication on his profile to show that the designer is not available on the platform Chat Feature Disabled: • Step: If the chat feature is temporarily disabled due to some technical issues. • System Response: The system displays an error message (e.g., "Chat feature is currently unavailable. Please try again later.") and gives an option to the user to get in contact with the designer via email.		

Name	Share Portfolio on Networking Platforms
Scope	Portfolio Publishing and Networking
Level	User goal
Actor	Primary Actor: Independent Designer Secondary Actor: Networking Platforms
Stakeholders and Interest	Designers: • Interest: To promote their portfolio work across different networking platforms (e.g Gmail, whatsapp and linkedin) to gain visibility and connect with big fashion companies Fabricae and Networking Platforms: • Interest: To increase the popularity of the platform so it can gain more exposure and designers can use it to share their designs
Precondition	• The designer must be logged in the platform • The designer must have an existing portfolio on the platform • The designer must have linked his account with the networking platform

Success guarantee	The designer will be able to share his portfolio with the networking platform of his choice with a link to his portfolio and a relevant caption	
Main success scenario		
	Actor Action	System Response
	1.The designer navigates to his own gallery and selects the published portfolio he wants to share	
		2.The system gives him list of his linked networking accounts options(whatsapp,email,linkedin etc)
	3. The designer selects the platform of his choice.	
		4. The system prompts the user to link their account first and redirects them to the specific platform login page.
	5.The user inputs his login information.	
		6.The system generates a shareable link and asks the user to input a caption relevant to the link
	7.The designer clicks on the ‘share’ button.	
		8.The system shares the portfolio link on the selected networking platforms and displays a confirmation message (e.g ‘ Your Portfolio has been successfully shared”)
Extensions	Networking Platform Unavailable:	
	• Step: When the selected networking platform is temporarily unavailable or connection fails. • System Response: The system displays an error message (e.g., "Unable to connect to the selected platform. Please try again later.")	

	and allows the designer to retry sharing once the platform is available. Invalid Social Media Authentication: • Step: When the Authentication with the platforms fail (e.g expired tokens) • System Response: The system displays a message to the user to re-authenticate with the platform Network Connectivity Issue: • Step: If the network connection fails while sharing portfolio • System Response: The system prompts an error message to the user that internet connection is not working and he needs to retry after a while
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2.1 Functional requirements

This section describes the functional requirements of the system. These requirements are organized by features.

2.2.1 Module 1

Following are the functional requirements for Module 1:

1. **FR1:** The system shall allow designers to generate textile patterns by entering text descriptions.
2. **FR2:** The system shall allow designers to save their generated patterns.
3. **FR3:** The system shall allow designers to view their generated patterns.

2.2.2 Module 2

Following are the functional requirements for Module 2:

1. **FR1:** The system shall allow designers to upload pencil sketches from their local machine.
2. **FR2:** The system shall allow designers to convert their uploaded pencil sketches into images.
3. **FR3:** The system shall allow designers to save the generated digital pattern in history.
4. **FR4:** The system shall allow designers to view the generated digital pattern from history.

2.2.3 Module 3

Following are the functional requirements for Module 3:

1. **FR1:** The system shall allow designers to edit the pattern by changing its colors.
2. **FR2:** The system shall allow designers to edit the pattern by changing the alignment such as rotate, scale motifs inside a pattern.
3. **FR3:** The system shall allow the designers to crop a subpart of the pattern.
4. **FR4:** The system shall allow designers to add filters on the edited patterns.

2.2.4 Module 4

Following are the functional requirements for Module 4:

1. **FR1:** The system shall allow designers to select 3D clothing models from the given options to test their patterns.
2. **FR2:** The system shall allow designers to drag and position the embellishments such as buttons and laces on top of the 3D model.
3. **FR3:** The system shall allow designers to view the 3D mock-up in 360-degree view.
4. **FR4:** The system shall allow designers to apply their customized patterns on the 3D clothing model.
5. **FR5:** The system shall allow designers to export their 3D designs in PNG, glTF and OBJ format.
6. **FR6:** The system shall allow designers to protect their patterns by applying watermark over them.

2.2.5 Module 5

Following are the functional requirements for Module 5:

1. **FR1:** The system shall provide the designer with the option to choose any one template from an option of two.
2. **FR2:** The system shall allow the designer to edit the portfolio template by adding text of his/her own choice.
3. **FR2:** The system shall allow the designer to edit the portfolio template by uploading pattern and model images from his saved patterns and models.
4. **FR2:** The system shall allow the designer to edit the text inside the portfolio template using bold, italic and alignment options.

2.2.6 Module 6

Following are the functional requirements for Module 6:

1. **FR1:** The system shall allow the designer to publish his/her portfolio on the platform.
2. **FR2:** The system shall allow the designer to edit his profile image and name.
3. **FR3:** The system shall allow the designer to add his generated and edited patterns in the design gallery.
4. **FR4:** The system shall allow the designer to share his portfolio via WhatsApp, LinkedIn and Gmail.
5. **FR5:** The system shall allow the designer to view notifications for message alerts.
6. **FR6:** The system shall allow the Industry professional to view the design gallery of the designers.
7. **FR7:** The system shall allow the Industry professional to view the portfolio of the designers.
8. **FR8:** The system shall allow the Industry professional to send text messages to the designer.
9. **FR9:** The system shall allow the designer to send text messages to the Industry professional once the Industry professional has initiated a chat.

2.3 Non-Functional Requirements

This section specifies non-functional requirements. These quality requirements are specific, quantitative, and verifiable.

2.3.1 Reliability

The system must save the in-progress edited pattern after every 10 minutes.

2.3.2 Usability

The system shall require less than 30 minutes to learn the basic functionalities of the app.

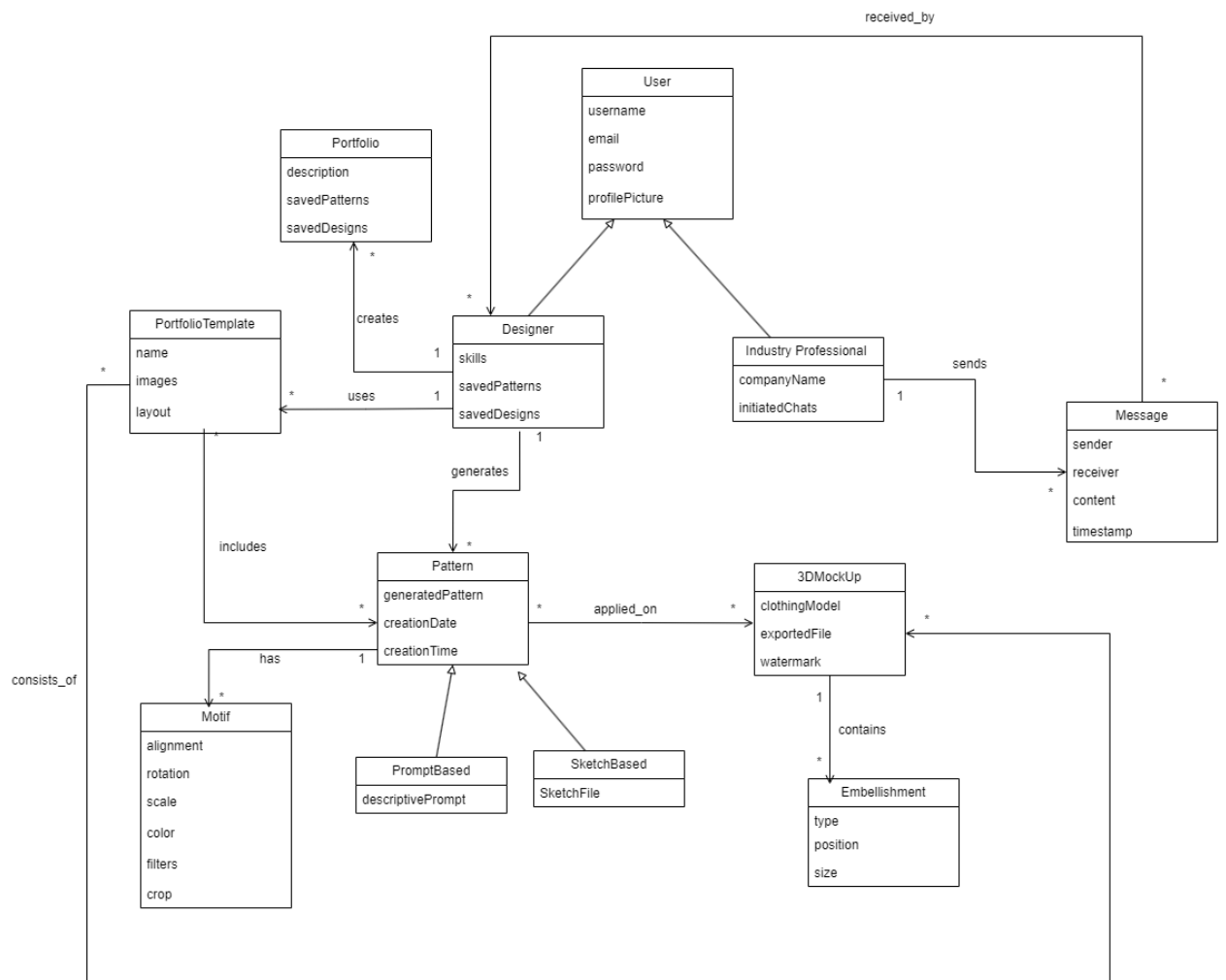
2.3.3 Performance

The system shall generate patterns within 2 minutes.

2.3.4 Security

The system shall ensure that the user data is encrypted.

2.4 Domain Model

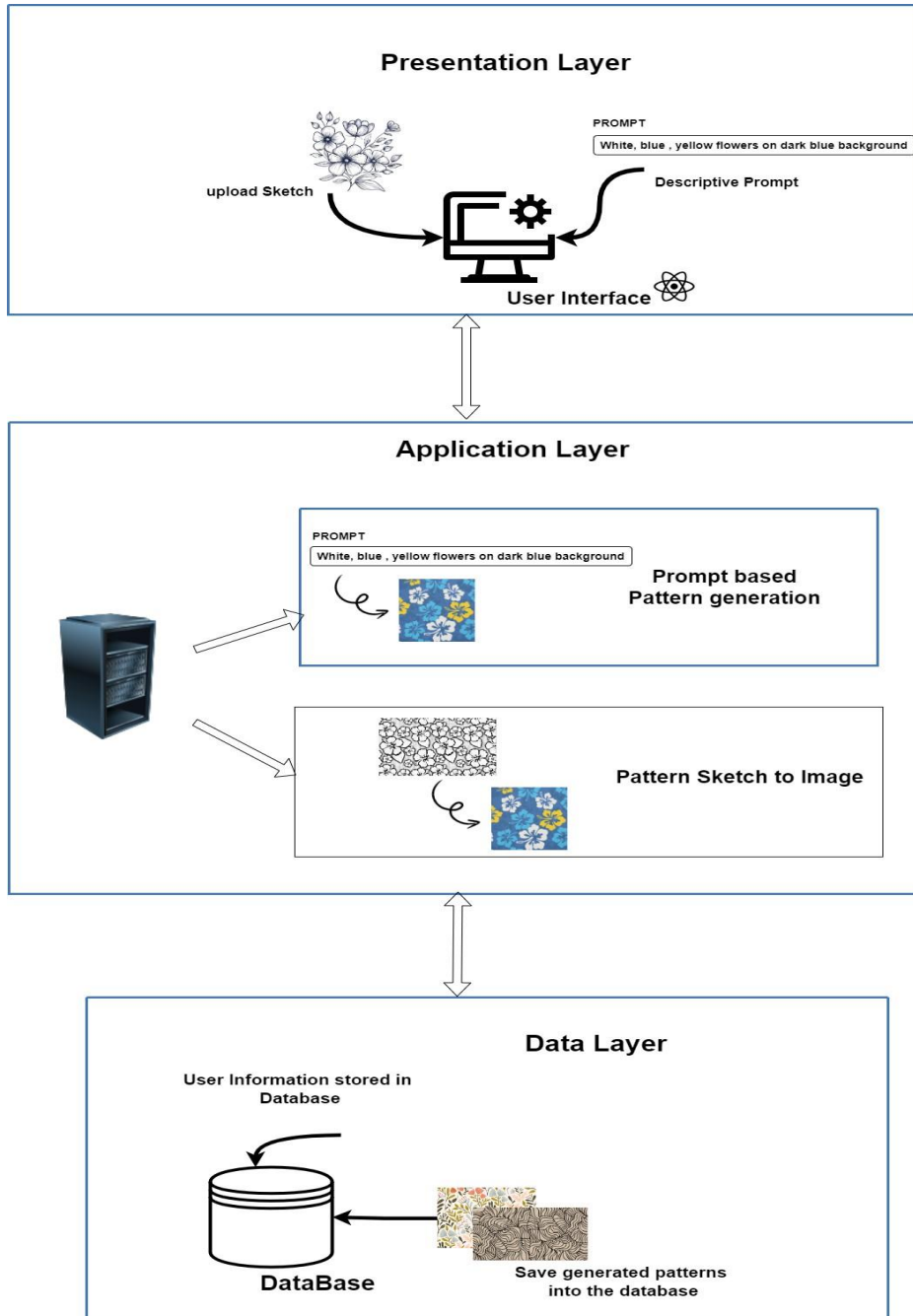


Chapter 3

System Overview

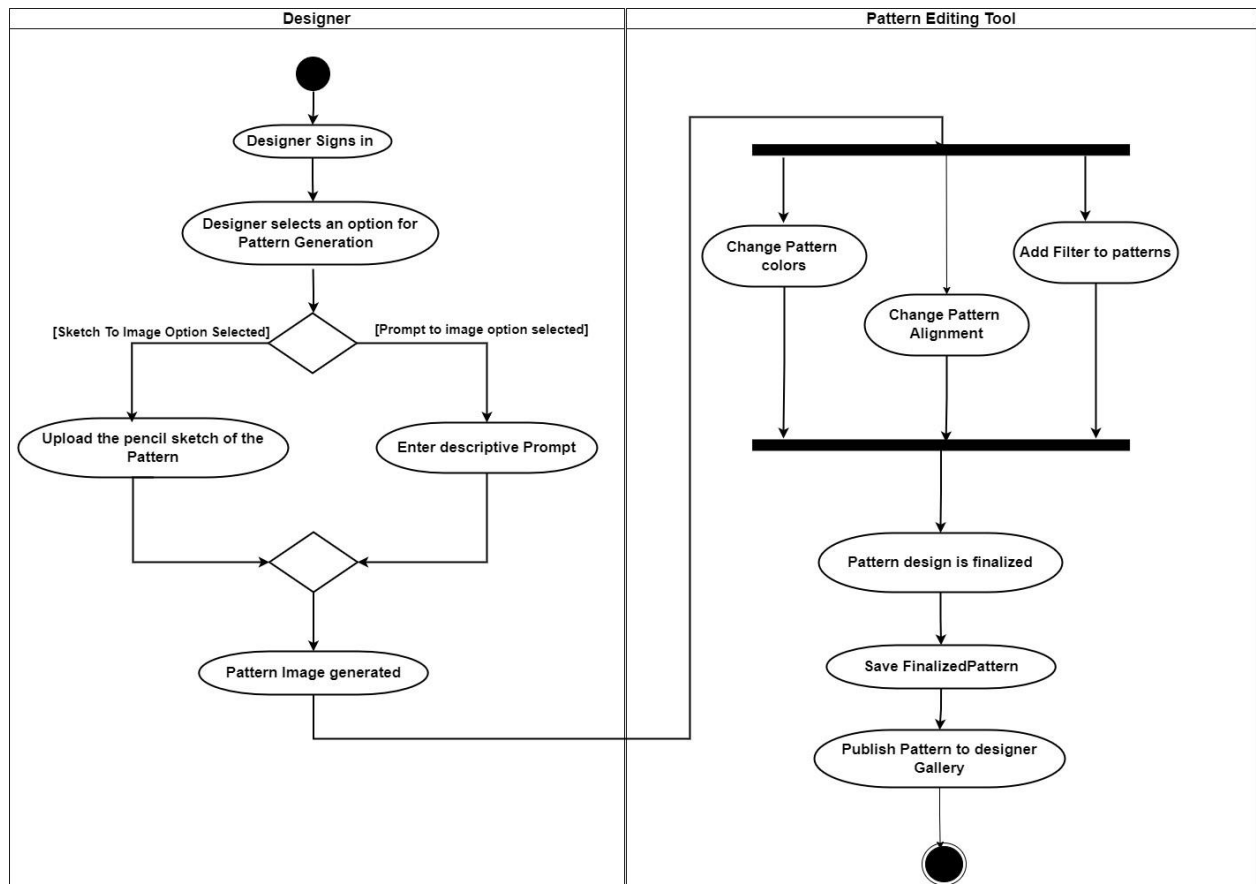
Fabricae is a web app that is designed to ease the process of pattern creation for the fashion designers. Moreover, it also provides an opportunity for these designers to connect with industry professionals by publishing their portfolios on our platform. The project is targeting the independent designers as well as small-scale fashion designers who are creative enough to transform their thoughts and sketches into realistic design patterns. Moreover, they are also given the opportunity to further edit and visualize their patterns on 3D mockups. The platform's architecture is a 3-Tier architecture for scalability and separation of concerns.

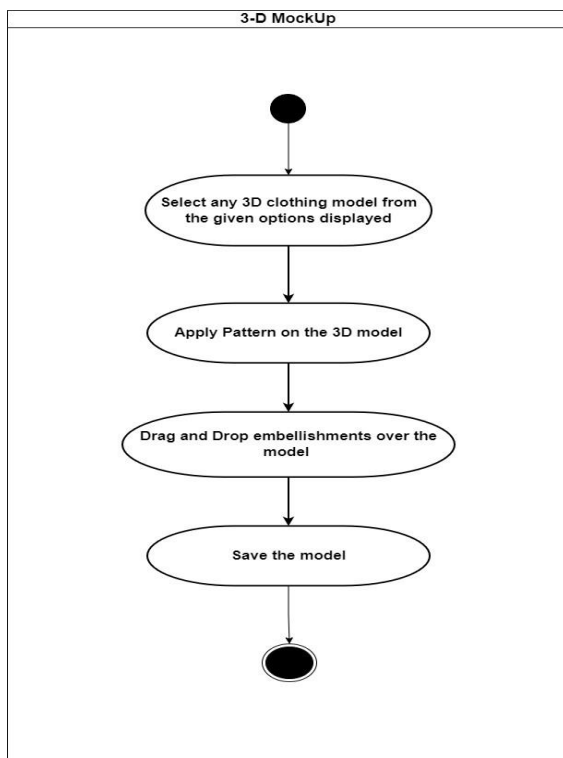
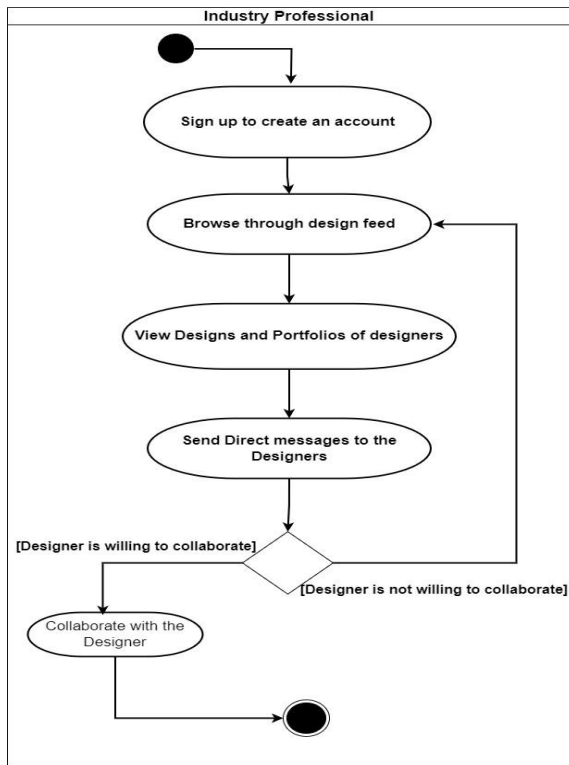
3.1 Architectural Design

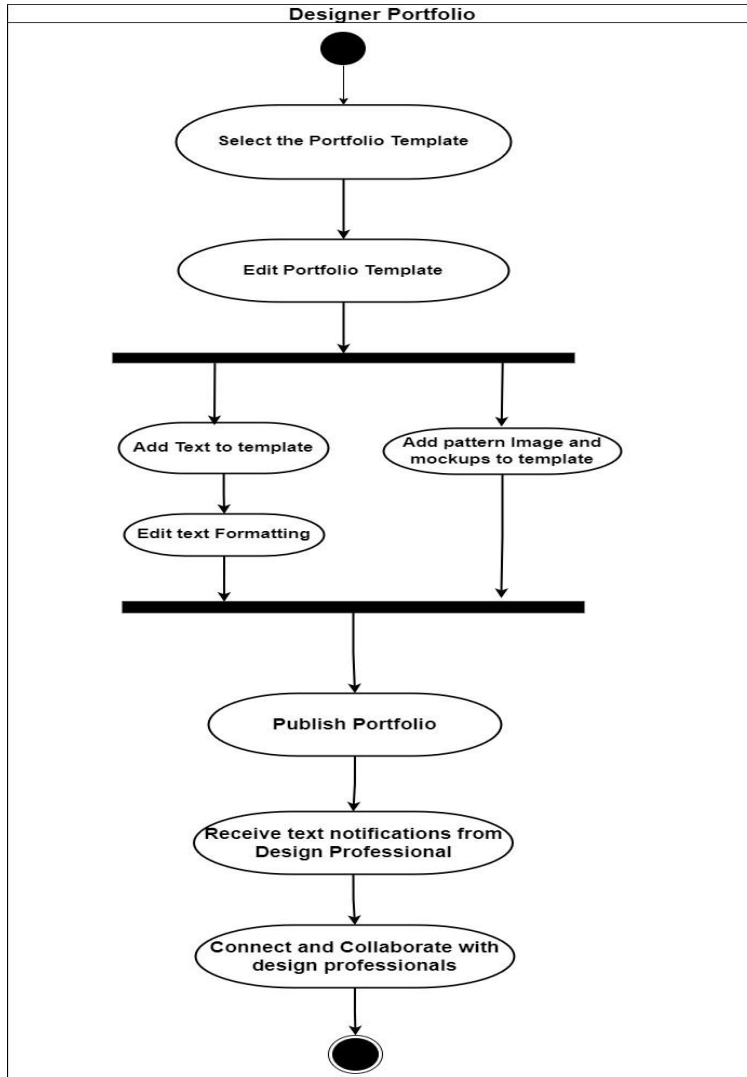


3.2 Design Models

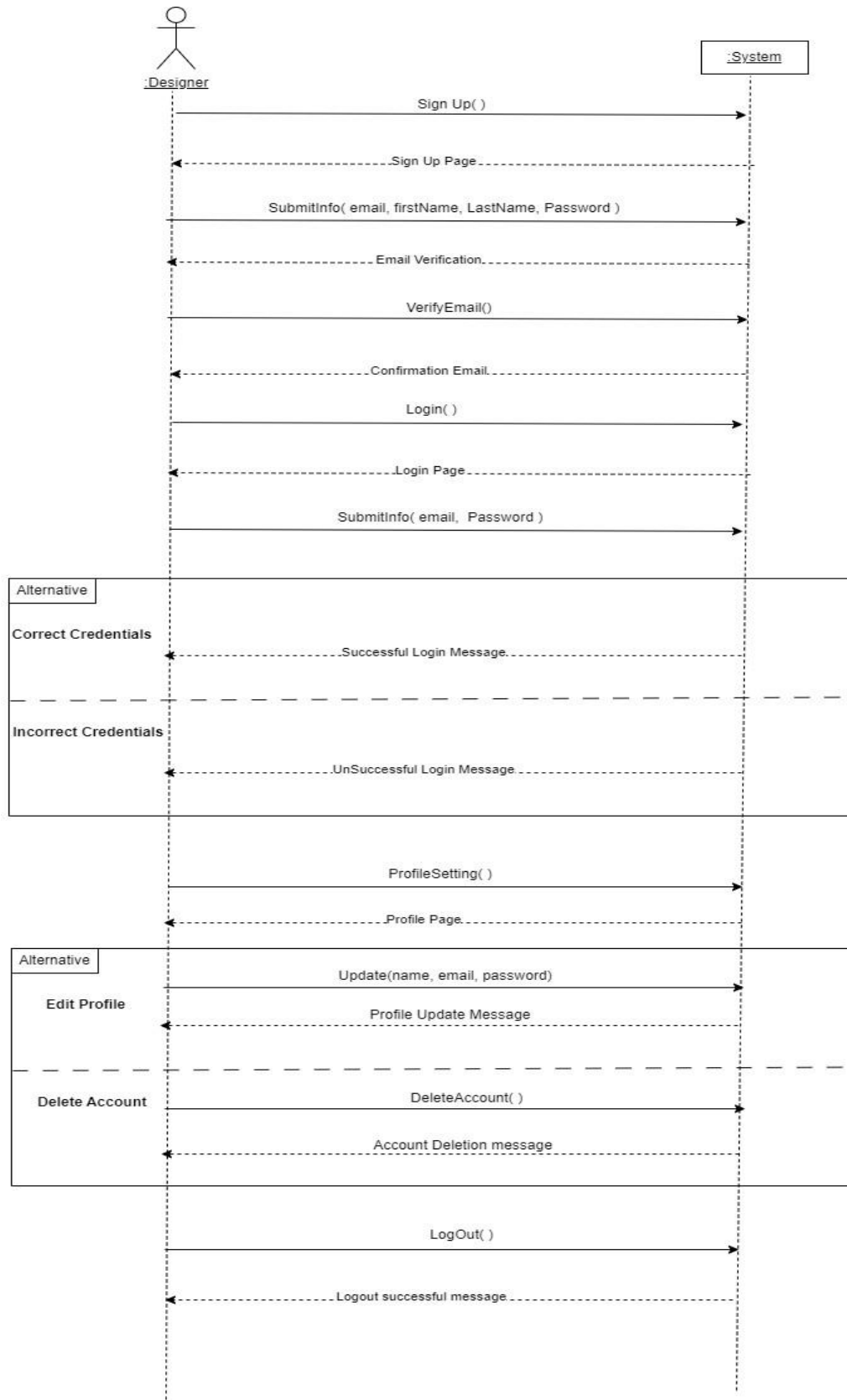
3.2.1 Activity Diagrams

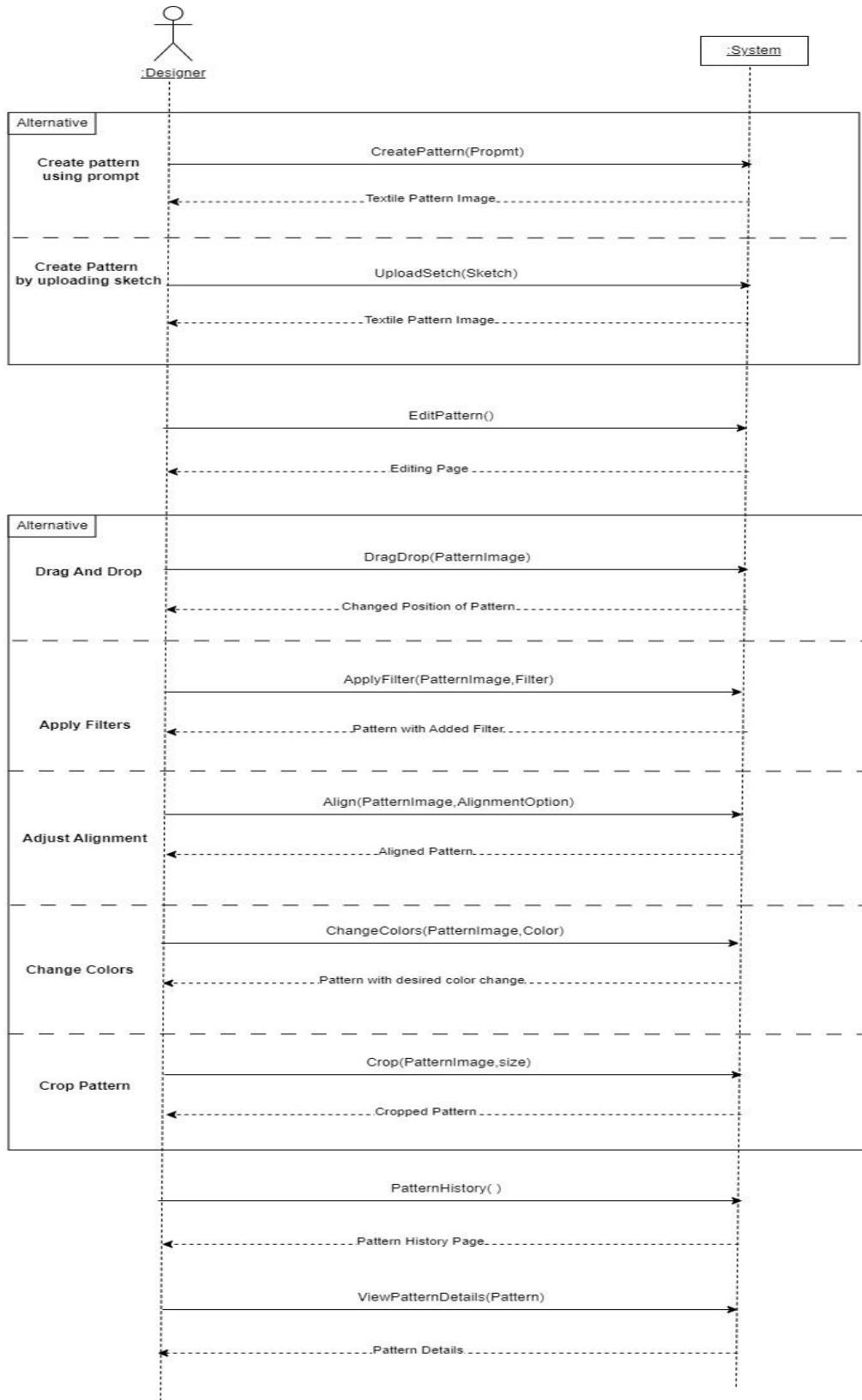


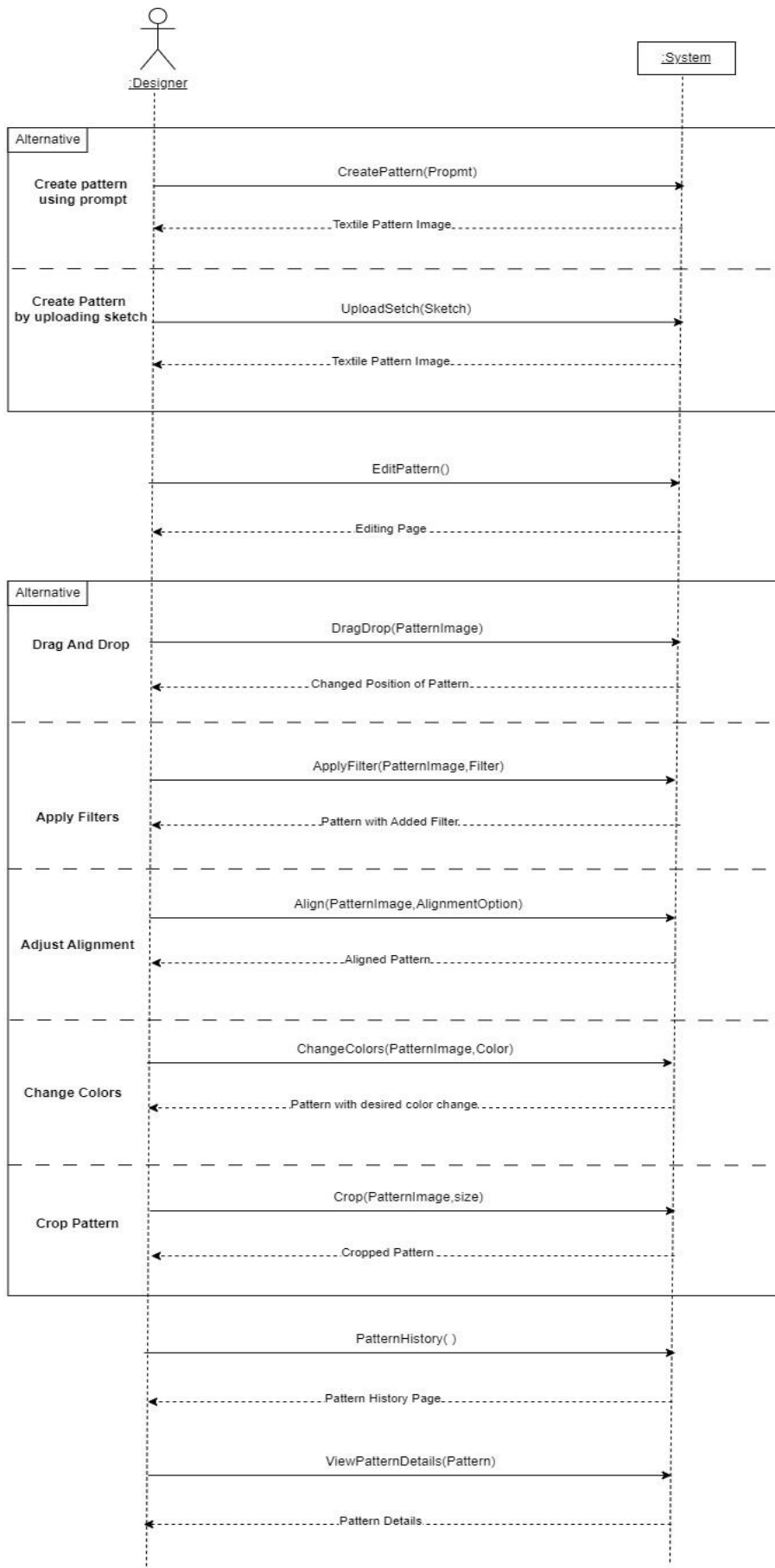


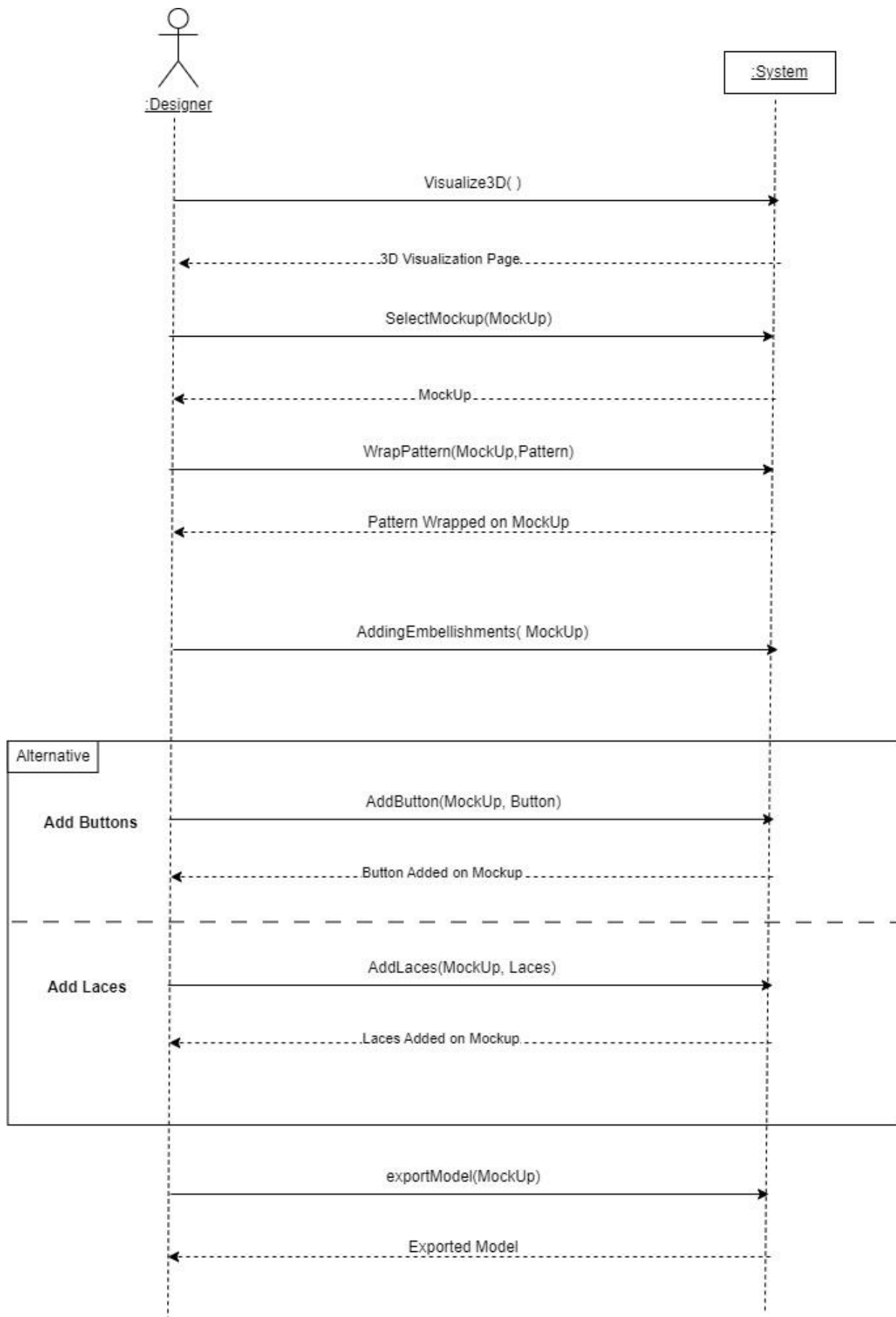


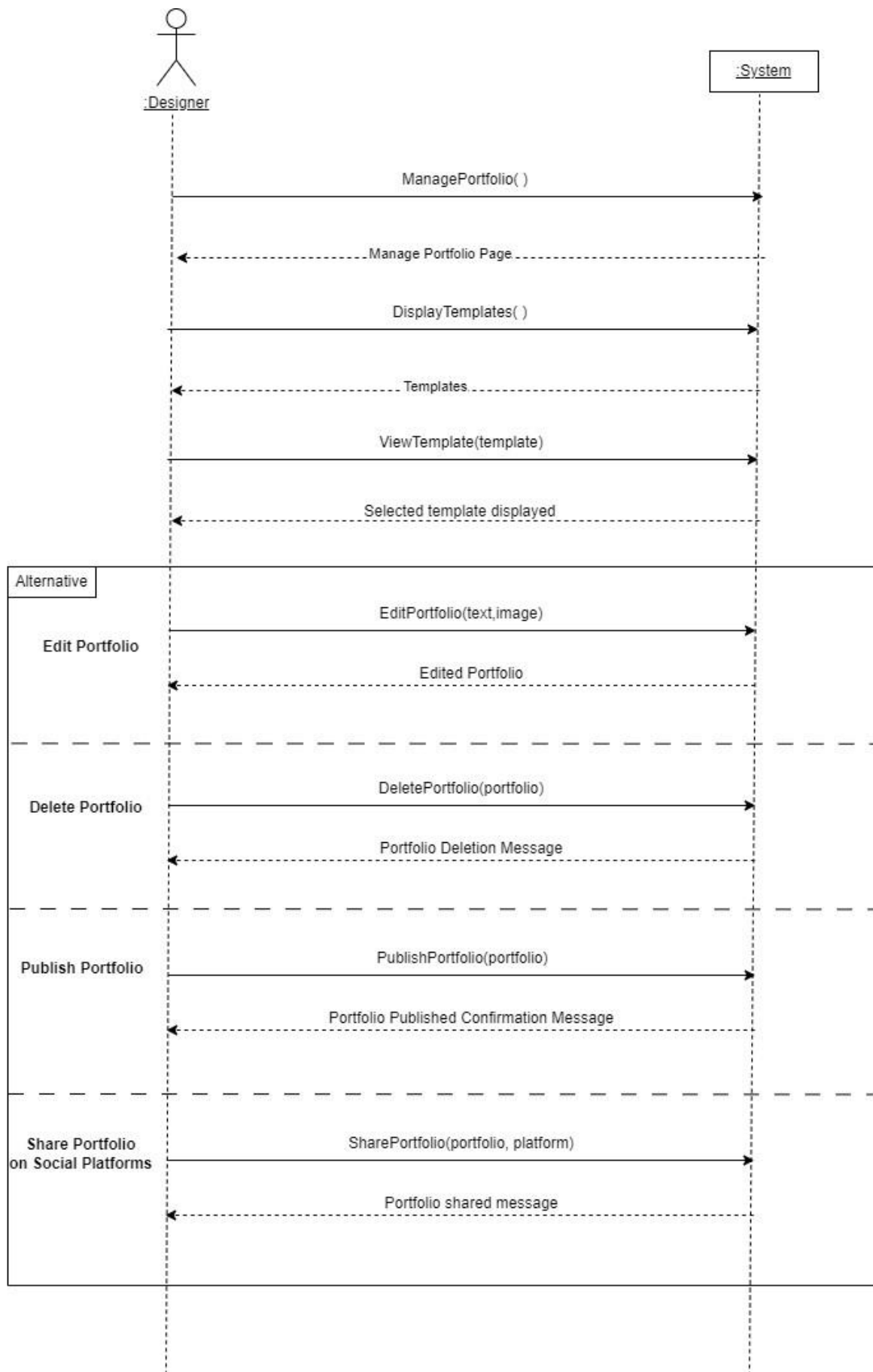
3.2.2 System Sequence Diagrams

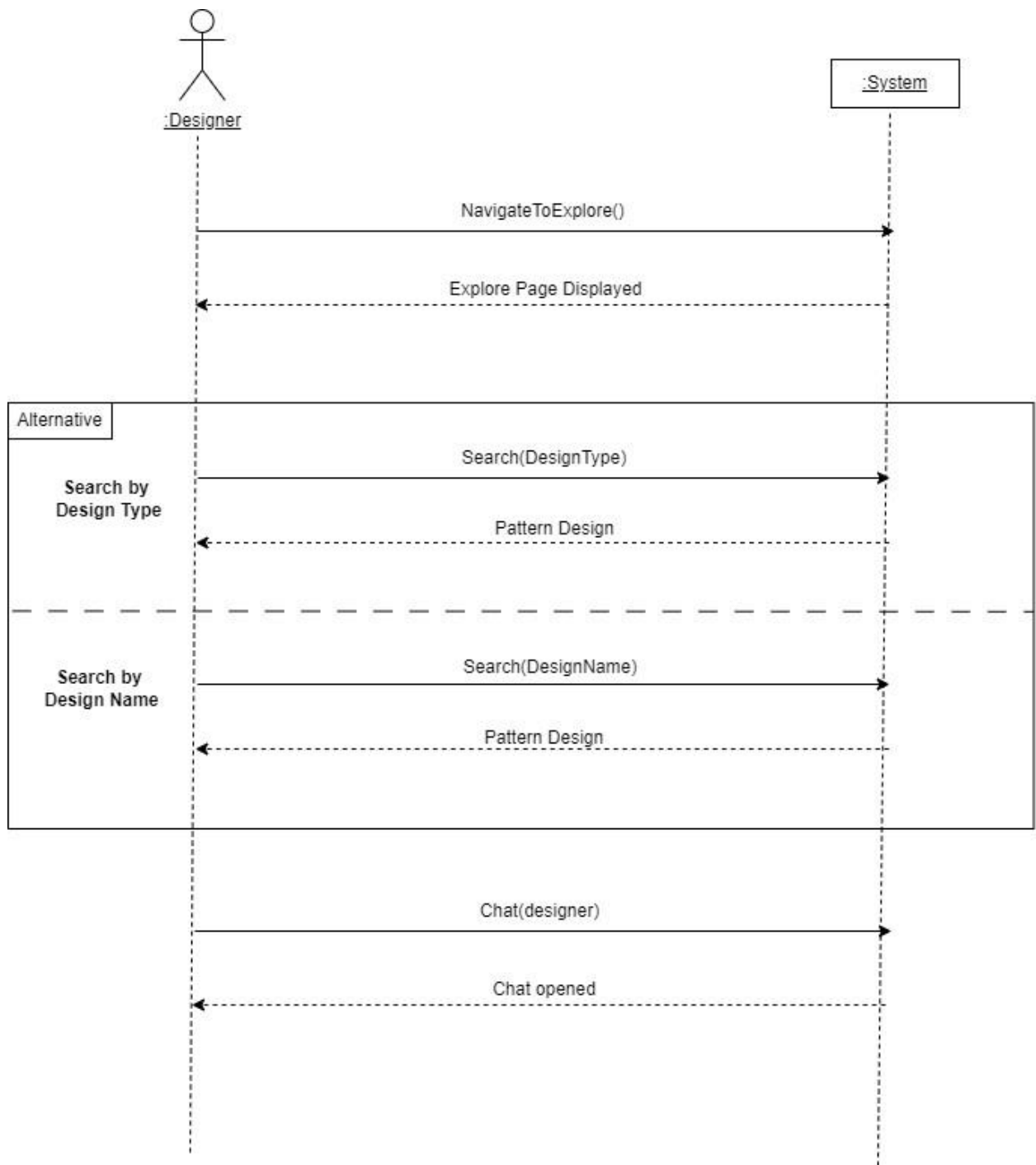






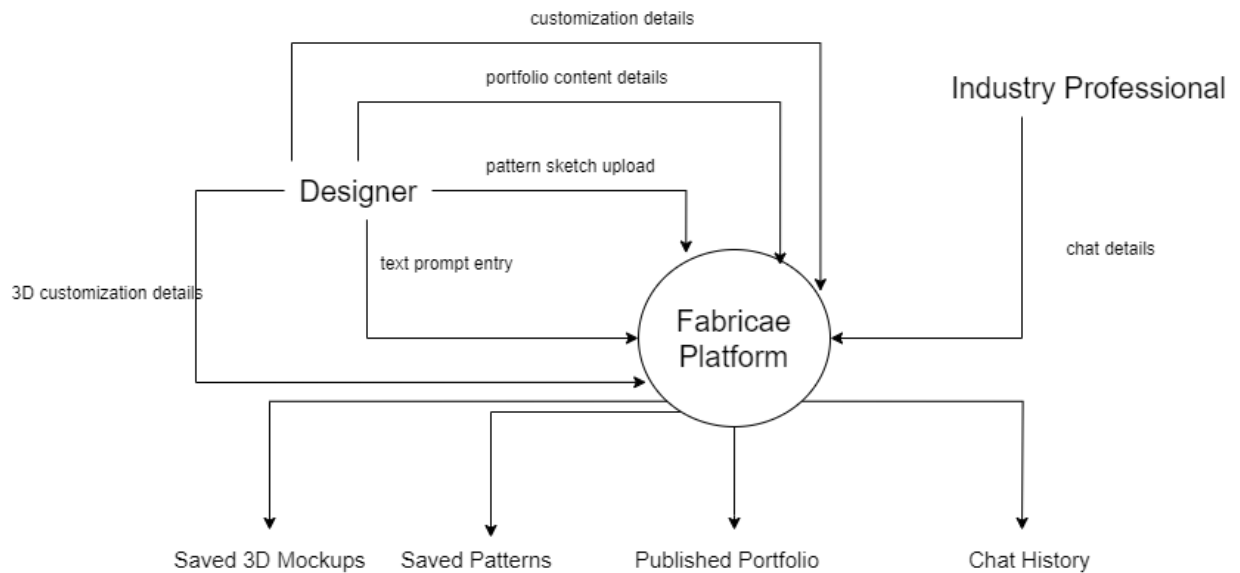




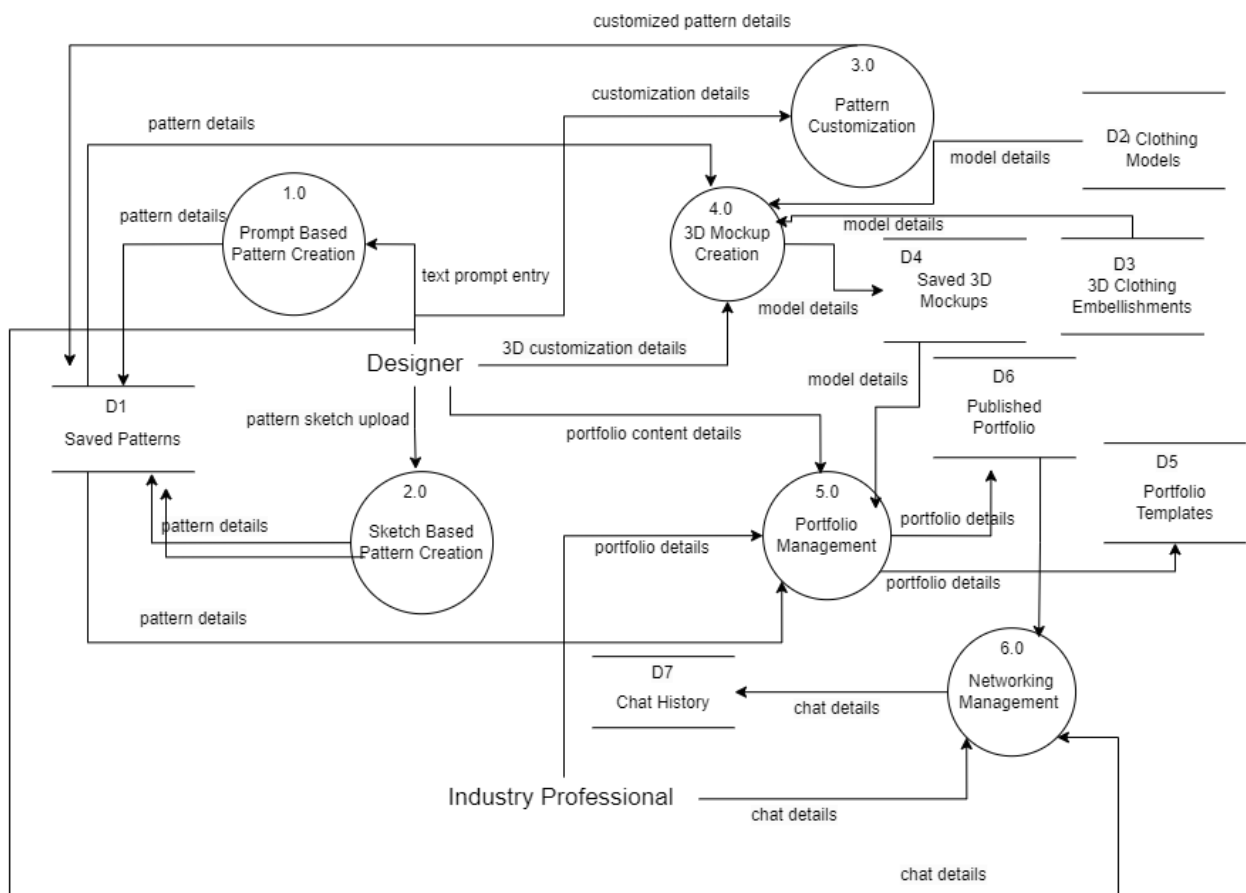


3.2.2 Data Flow Diagram

Level 0

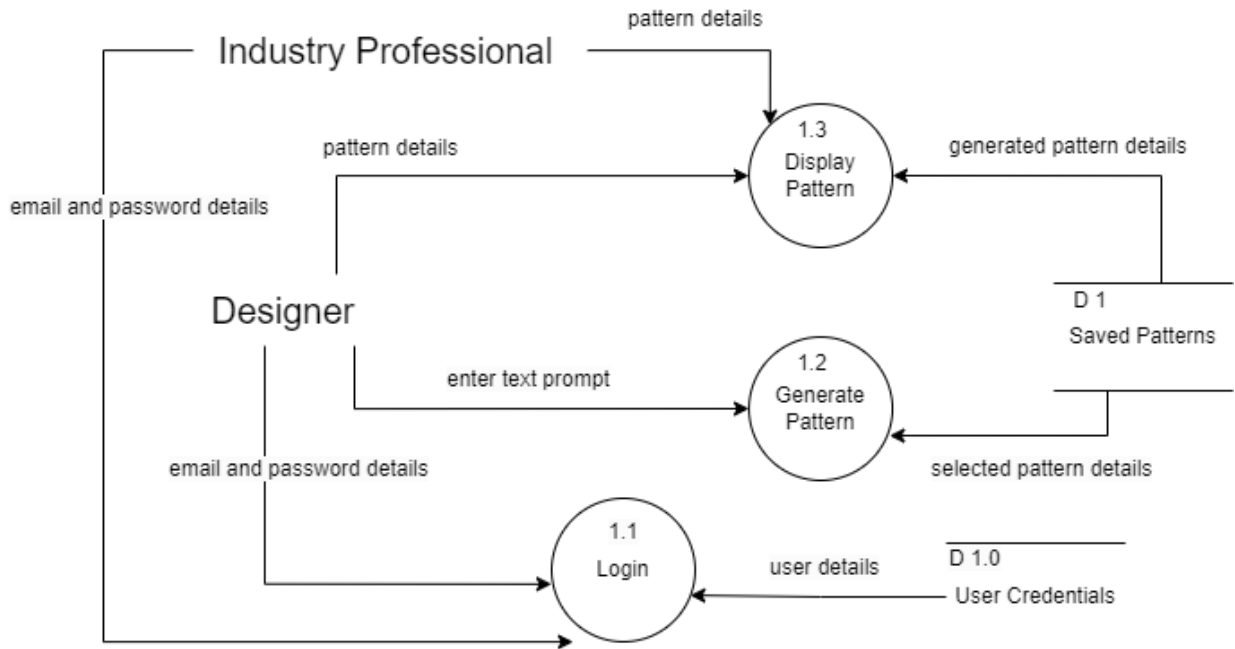


Level 1

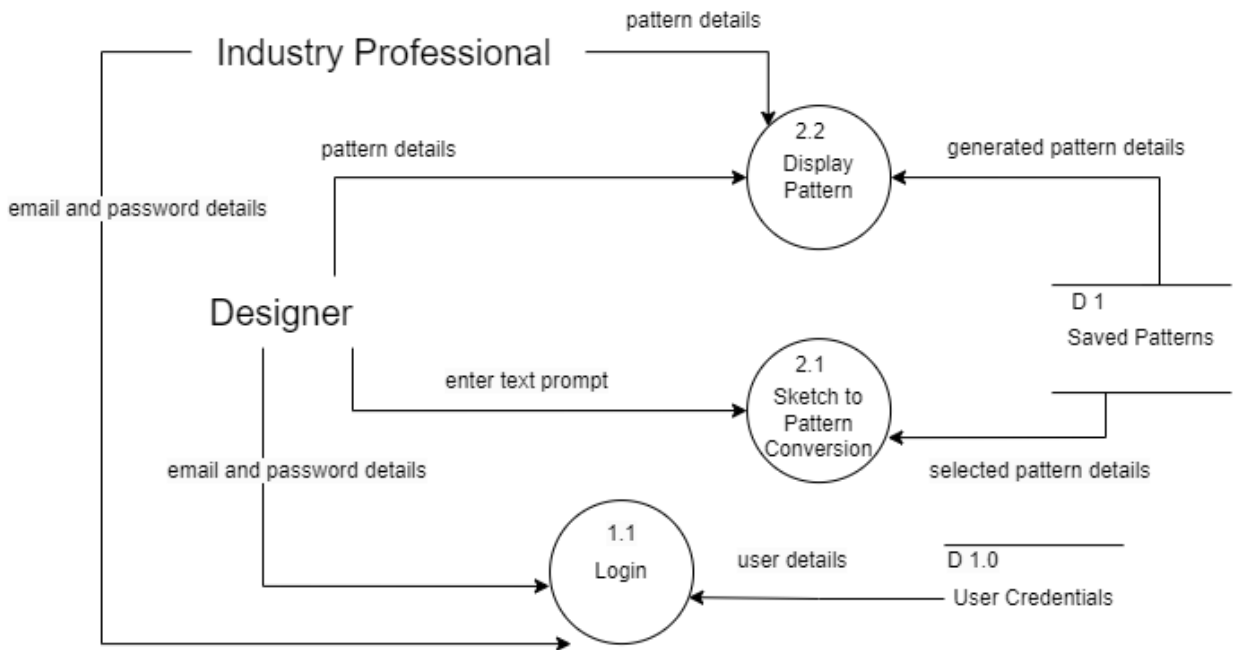


Level 2

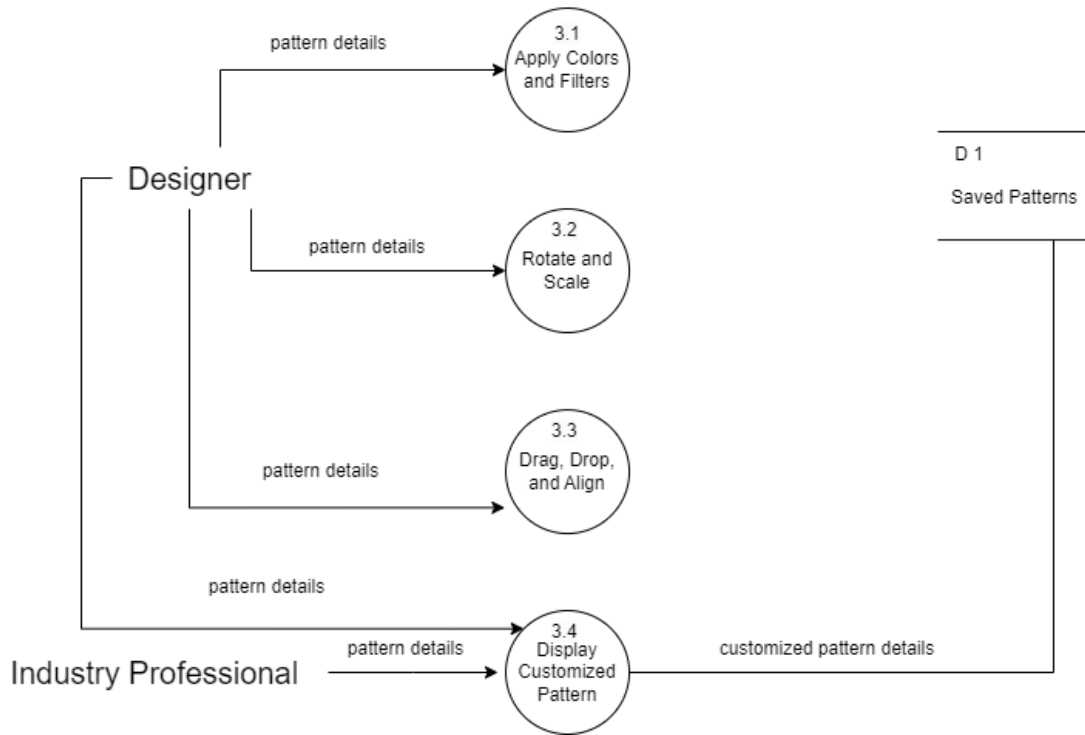
Prompt Based Pattern Creation



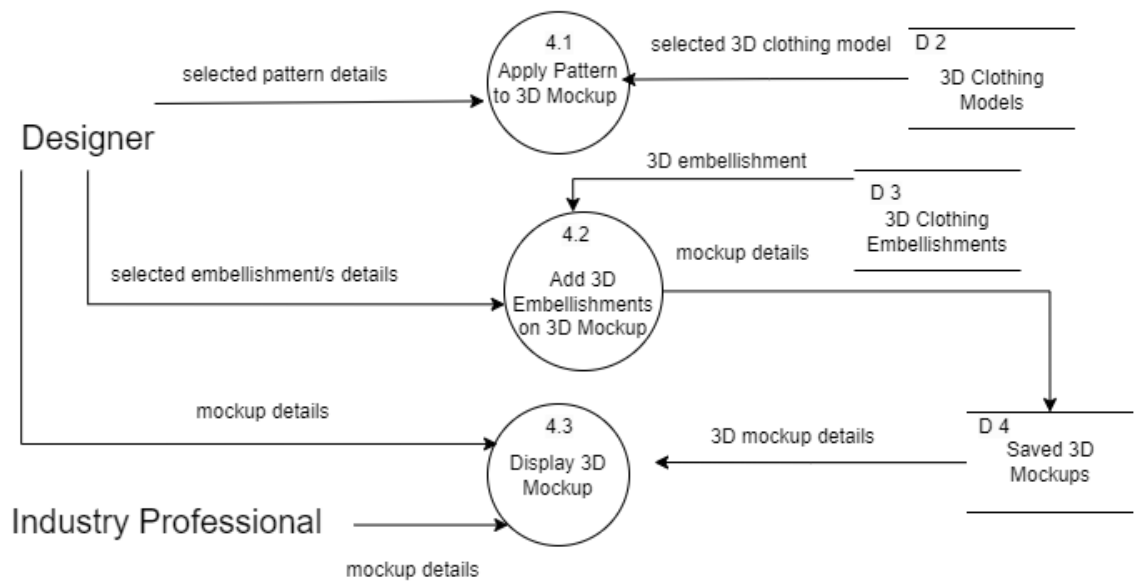
Sketch Based Pattern Creation



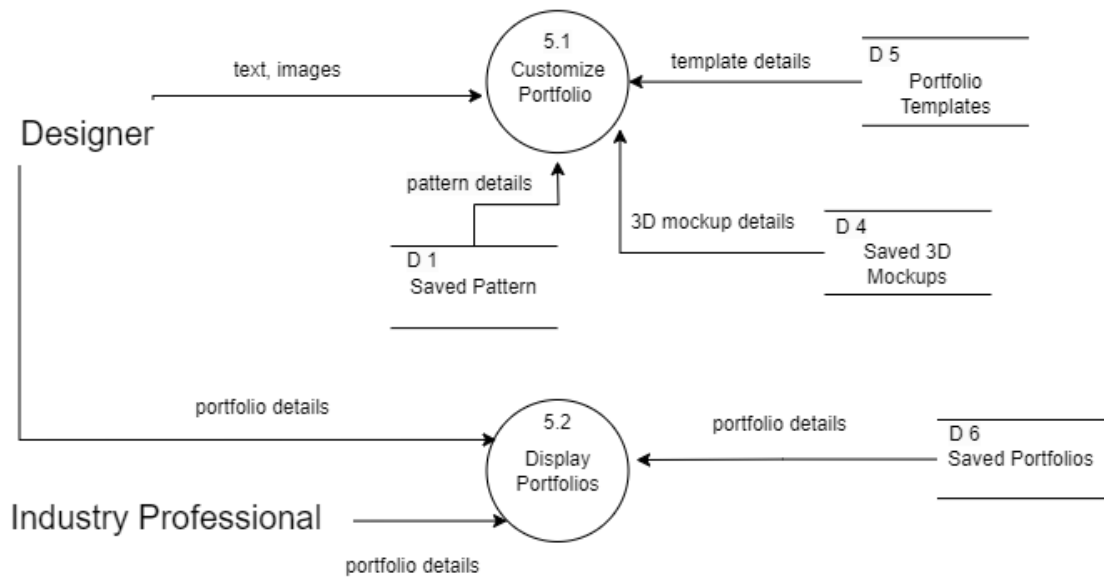
Pattern Customization



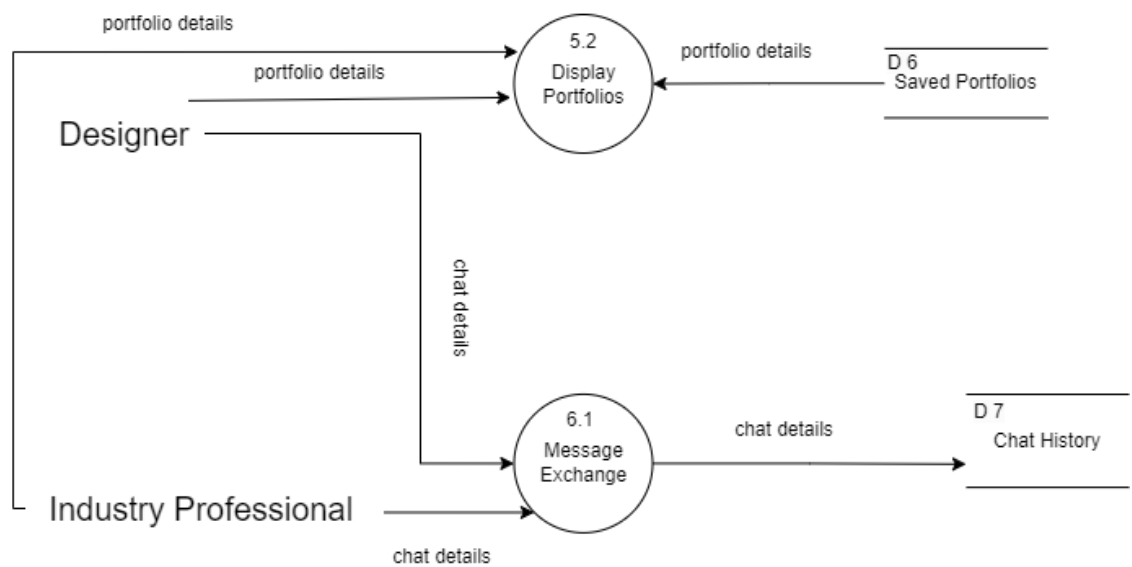
3D Mockup Creation



Portfolio Management



Networking Management



3.3 Data Design

```
"user":
{
  "userId": "string",
  "username": "string",
  "email": "string",
  "password": "string",
  "role": "enum", // "Designer" or "IndustryProfessional"
  "profilePictureUrl": "string",
  "createdAt": "date",
  "updatedAt": "date",
  "Designer":
  {
    "skills": ["string"],
    "portfolioId": "string",
    "savedPatterns": ["patternId"],
    "savedDesigns": ["mockupId "]
  },
  "IndustryProfessional":
  {
    "companyName": "string",
    "initiatedChats": ["chatId"]
  }
},
```

```
"motif":
{
  "motifId": "string",
  "alignment": "string",
  "rotation": "number",
  "scale": "number",
  "color": "string",
  "crop": "string",
  "filters": ["string"]
}
```

```

"pattern":
{
  "patternId": "string",
  "designerId": "userId",
  "createdAt": "date",
  "patternType": "enum", // "PromptGenerated" or "SketchUploaded"
  "patternData": {
    "prompt": "string", // Only for PromptGenerated
    "sketchImageUrl": "string", // Only for SketchUploaded
    "motifs": [motifId]
  }
}

```

```

"3dMockup":
{
  "mockupId": "string",
  "pattern": " patternId ",
  "garmentType": "enum", // "Shirt", "Pant", "Dress", "Shalwar Kameez", etc.
  "embellishments": [embellishmentId],
  "exportOptions": {
    "format": "enum", // "PNG", "OBJ"
    "watermarked": "boolean"
  }
},

```

```
"embellishment":  
{  
  "embellishmentId": "string",  
  "embellishmentType": "enum", // "Button", "Lace"  
  "position": "string",  
  "size": "string"  
},
```

```
"portfolioTemplates":  
{  
  "templateId": "string",  
  "templateName": "string",  
  "layout": "string",  
  "allowedContent": ["string"] // "Text", "Image"  
}
```

```
"portfolio":  
{  
  "portfolioId": "string",  
  "designer": "userId",  
  "template": " templateId ",  
  "createdAt": "date",  
  "updatedAt": "date",  
  "title": "string",  
  "description": "string",  
  "savedDesigns": [mockupId],  
  "savedPatterns": [patternId],  
  "isPublished": "boolean",  
},
```

"message":

```
{  
  "messageId ": "string",  
  "messageText": "string",  
  "timestamp": "date"  
},
```

"chat":

```
{  
  "chatId ": "string",  
  "sender": "userId",  
  "receiver": "userId",  
  "messages": [messageId]  
},
```

